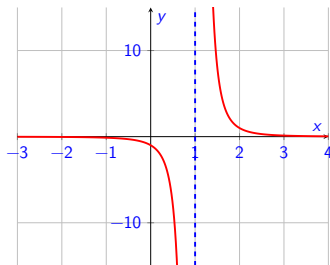


# Week 03, Lecture 2 Infinite and one-sided limits; Vertical Asymptotes

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*This slides are by Niall Madden, with some content based on notes by Dr Kirsten Pfeiffer, and from Strang & Herman's "Calculus".*

### Reminders

- ▶ **Problem Set 1** due 5pm, Monday 28 September. You may access it multiple times, by clicking on [Assignments ... Problem Set 1 ...](#) and then, at the bottom of the page:  
[Load Problem Set 1 in a new window](#)
- ▶ The **Tutorial Sheet** is available from Canvas.
- ▶ Problem Set 2 will open by Friday.
- ▶ The first (of two) class tests will take place Tuesday, 6th October.  
If you wish to avail of Reasonable Accommodations for it, please complete this form:  
<https://forms.cloud.microsoft/e/rBiq8RHHY2> by **10am, Tuesday 29 September**.

# Today - a one-sided view of the following topics:

- 1 Infinite Limits
- 2 Digression: How fast can an object travel?
- 3 One-sided Limits
  - Notation
- 4 Piecewise functions
  - Empty and full circle notation
  - Existence of a limit
- 5 Vertical Asymptotes
  - Horizontal Asymptotes
- 6 Exercises

For more, see Chapter 2 (Limits) in **Calculus** by Strang & Herman. See [openstax.org/books/calculus-volume-1/pages/2-introduction](https://openstax.org/books/calculus-volume-1/pages/2-introduction)

In particular §2.2: One-sided limits (§2.2.4+§2.2.5) and vertical asymptotes (§2.2.7).

# Infinite Limits

So far, we've had lots of examples that are a little like:

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2 - x + 1}{(x - 1)^2} = 2.$$

(Check that this is correct).

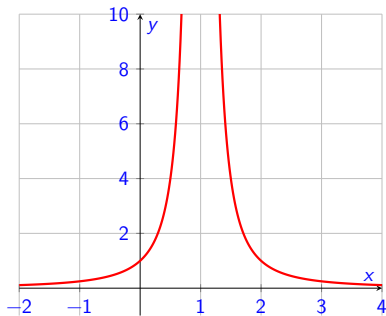
But what about

$$\lim_{x \rightarrow 1} \frac{1}{(x - 1)^2} = ???$$

Let's plot it and see:

# Infinite Limits

$$f(x) = \frac{1}{(x-1)^2}$$



As  $x$  get closer and closer to 1, the value of  $f(x)$  gets larger and larger. In fact, it becomes infinite.

For this we write

$$\lim_{x \rightarrow 1} f(x) = \infty.$$

## Digression: How fast can an object travel?

- ▶ Q: Is there any limit to the speed at which an object can travel?
- ▶ A: Yes! (Assuming you believe Einstein)

Thanks to Einstein ( $E = mc^2$ ), Lorenz and others, it is known that the mass of a moving charged particle behaves like

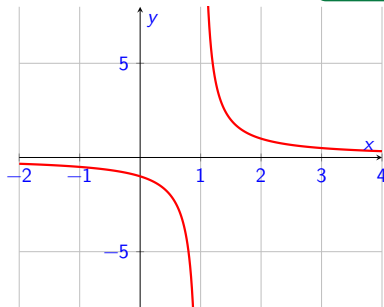
$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where  $m_0$  is its mass at rest,  $c$  is the speed of light, and  $v$  is the particles current speed. What happens as  $v \rightarrow c$ ?

# One-sided Limits

Let's consider a motivating example, very similar to the one where we introduced  $\infty$ .

$$f(x) = \frac{1}{x-1}$$



As  $x$  get closer and closer to 1, then  $f(x) \rightarrow -\infty$  or  $f(x) \rightarrow \infty$ , depending on whether  $x$  approaches 1 from the left or right.

To express this, we need the concept of a **one-sided limit**

$\lim_{x \rightarrow a^-} f(x)$  is: **limit of  $f$  as  $x$  approaches  $a$  from the left**

$\lim_{x \rightarrow a^+} f(x)$  is: **limit of  $f$  as  $x$  approaches  $a$  from the right**

These mean that

- ▶ if  $\lim_{x \rightarrow a^-} f(x) = L$ , then we can make  $f(x)$  as close to  $L$  as we would like by taking  $x < a$  as close to  $a$  as needed.
- ▶ If  $\lim_{x \rightarrow a^+} f(x) = L$ , then we can make  $f(x)$  as close to  $L$  as we would like by taking  $x > a$  as close to  $a$  as needed.

$\lim_{x \rightarrow a^-} f(x)$  is: **limit of  $f$  as  $x$  approaches  $a$  from the left**

$\lim_{x \rightarrow a^+} f(x)$  is: **limit of  $f$  as  $x$  approaches  $a$  from the right**

In the previous example, with  $f(x) = \frac{1}{x-1}$ , we have

►  $\lim_{x \rightarrow 1^-} f(x) = -\infty$

►  $\lim_{x \rightarrow 1^+} f(x) = \infty$

# Piecewise functions

In many important examples, we encounter functions that have different definitions in different regions. The most classic example is the **absolute value function**:

$$|x| = \begin{cases} -x & \text{for } x < 0 \\ x & \text{for } x \geq 0. \end{cases}$$

Care has to be taken when evaluating the limits of such functions....

# Piecewise functions

## Example

Sketch the function

$$f(x) = \begin{cases} 3 - x, & \text{if } x \leq 2 \\ \frac{x}{2} + 1, & \text{if } x > 2 \end{cases}$$

Find  $\lim_{x \rightarrow 2^-} f(x)$  and  $\lim_{x \rightarrow 2^+} f(x)$ .

### *Empty and Full Circle Notation:*

In the previous sketch, we use the convention that

- ▶ If the end point of a line segment is **not** included in its definition, it terminates with an **open circle**,  $\circ$
- ▶ If the end point of a line segment **is** included in its definition, it terminates with an **closed circle**  $\bullet$ .

# Existence of a limit

## Existence of a limit

$\lim_{x \rightarrow a} f(x)$  **exists** if and only if

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

So if  $\lim_{x \rightarrow a} f(x) = L$  exists, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

**though it is not necessary that**  $f(a) = L$

# Existence of a limit

## Example

Sketch the function

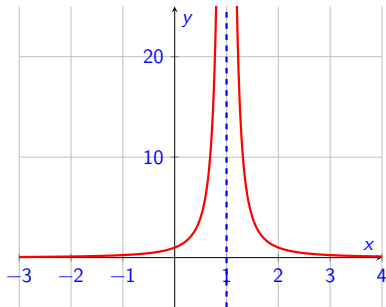
$$f(x) = \begin{cases} 3 - x, & x < 2 \\ 4, & x = 2 \\ \frac{x}{2}, & x > 2 \end{cases}$$

Determine if  $\lim_{x \rightarrow 2} f(x)$  exists.

# Vertical Asymptotes

Let's revisit an example from earlier:

$$f(x) = \frac{1}{(x-1)^2}$$



Note that the points on the graph having  $x$ -coordinates very near to 1 are very close to the vertical line  $x = 1$ . That is, as  $x$  approaches 1, the points on the graph of  $f(x)$  are closer to the line  $x = 1$ .

We call the line  $x = 1$  a **vertical asymptote** of the graph.

# Vertical Asymptotes

## Definition: Vertical Asymptote

The vertical line  $x = a$  is a **vertical asymptote** of  $f(x)$  if any of  $\lim_{x \rightarrow a^-} f(x)$ ,  $\lim_{x \rightarrow a^+} f(x)$ , or  $\lim_{x \rightarrow a} f(x)$  are  $\infty$  or  $-\infty$ .

To find a vertical asymptote of a function  $f(x) = \frac{p(x)}{q(x)}$ , we find a value,  $a$  for which  $p(a) \neq 0$  but  $q(a) = 0$ .

# Vertical Asymptotes

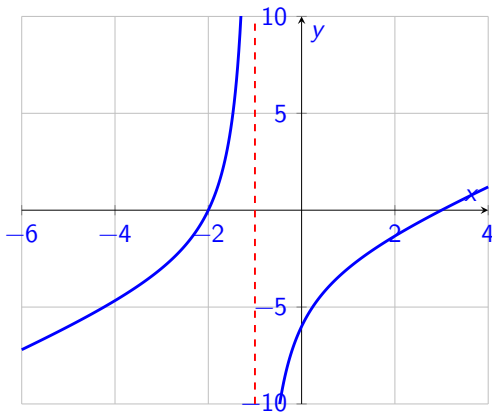
## Example

Find any vertical asymptotes of

$$f(x) = \frac{x^2 - x - 6}{x + 1}$$

# Vertical Asymptotes

$$f(x) = \frac{x^2 - x - 6}{x + 1}$$



# Vertical Asymptotes

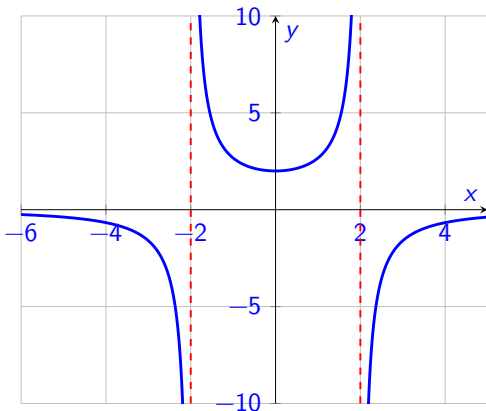
## Example

Find all vertical asymptotes of the graph of

$$g(x) = -\frac{8}{x^2 - 4}.$$

# Vertical Asymptotes

$$f(x) = -\frac{8}{x^2 - 4}$$



There is a related concept of a **horizontal asymptote**, but we'll save that for later, when we cover “limits at infinity”.

### Exercise 3.2.1

Find all the vertical asymptotes of  $f(x) = \frac{x+2}{x^2+2x-8}$ .

For more practice try the Exercises at the end of <https://openstax.org/books/calculus-volume-1/pages/2-3-the-limit-laws>, especially Problems 93–103.